

Hannah Liu

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Education

University of Waterloo

Bachelor of Software Engineering, Honours Co-op

Sep 2024 – Apr 2029

Waterloo, ON

Technical Skills

Languages: TypeScript/JavaScript, Python, C/C++, HTML/CSS, Ruby/Ruby on Rails, Java, SQL, Node.js

Frameworks/Tools: React/React Native, Git/GitHub, Figma, Jest, CI/CD, Playwright, Prometheus/Grafana, Chromatic, Storybook

Work Experience

Shopify *Top-tier performance rating*

Jan 2026 – Apr 2026

Software Engineering Intern

Toronto, ON

- Secured and scaled feature-access for **2+ million merchants** by engineering a **scalable feature-gating system** for Shopify Analytics; migrated legacy plan-based logic to a **centralized access control system** in **Ruby on Rails** to improve system auditability and monetization flexibility
- Empowered **2+ million merchants** with real-time business intelligence by engineering a high-scale product status filtering engine; optimized data ingestion by refining **Apache Beam ETL data pipelines** and authoring complex **ClickHouse schema migrations** to ensure high-performance analytical accuracy
- Accelerated Shopify platform performance for global users by architecting **10+ high-performance Preact web components**; reduced initial loading time by **82%** and reduced core application weight by **90%** by migrating from legacy React components while evaluating **60+** pull requests to ensure elite quality
- Modernized data accessibility for **1.15 million active merchants** by leading the end-to-end development of the Shopify Analytics data navigation system; engineered a platform-wide standard for temporal data filtering using **React/Preact** and facilitated stakeholder feedback loops to ensure a consistent, world-class user experience across the ecosystem

Shopify

May 2025 – Aug 2025

Software Engineering Intern

Toronto, ON

- Increased test reliability from **30% to 99%** by writing and maintaining **end-to-end, unit and integration** tests using **Jest** and **Playwright**, monitoring for failures using an internal **Prometheus and Grafana** platform
- Engineered interactive UI features for the Shopify Analytics Dashboard used by **2+ million** Shopify merchants, optimizing consumer insights through **complex state management** for features like draggable metrics and animated transitions
- Integrated a widely used table component using **TypeScript, Polaris, and CSS** in order to generate the **Customer Cohort Analysis** report used by **2+ million** Shopify merchants while aligning with **modern architectural best practices**
- Enhanced accessibility of the Shopify Admin Dashboard by adding **ARIA labels** and testing with **VoiceOver**, aligning with **WCAG 2.1 standards** to support screen reader users

Projects

DoorSense | React Native, Expo, TypeScript, CSS



- Collaborated in a team of six to build an embedded smart security system consisting of a mobile **iOS** app
- Designed and developed the mobile **iOS** application using **React Native, Expo, TypeScript, and CSS**, integrating real-time camera monitoring and an automated notification system for parcel detection and suspicious activity alerts
- Filmed and edited the demo video using **Canva** to present all features of the product to **140+** classmates

ShopSmart | React Native, TypeScript, CSS, Expo



- Led the process of designing and developing ShopSmart, an **iOS** mobile app that generates optimized grocery shopping routes based on a list of items, factoring in prices, travel time, and store proximity
- Built using **React Native (with Expo), TypeScript, and CSS**, focusing on responsive UI

Reflect. | React Native, TypeScript, Expo, CSS



- Designed and built Reflect., an **iOS** social media platform where users can record and share their accomplishments and goals to fuel motivation and promote mental health awareness
- Used **Figma** for designing, **React Native + Expo** for development, **TypeScript + CSS** for front-end

Undead Uprising | Python/Pygame, OOP, Figma



- Led a team of six in developing Undead Uprising, a zombie apocalypse video game, which was designed and implemented using **Figma**, Pixilart and **Pygame (Python)**, enjoyed by **30+** users